Airlock

A household agent that pays your recurring bills on its own but can't be talked into the wrong one — prompt-injection-resistant by construction, on ASI:One.

Fetch-A-Thon 3.0 · ASI:One Agent Challenge: From Intent to Action

Team: <Team name>

• <Name - role - email/GitHub>

1. Submission links

Deliverable	Link
ASI:One shared chat (complete workflow)	https://asi1.ai/shared-chat/<id>
Public GitHub repository	https://github.com/<you>/airlock
Demo video	https://youtu.be/<id>
Agentverse profile: Airlock (orchestrator)	https://agentverse.ai/agents/details/agent1qvlcnk524vntglwps897qf0dst0uqzcznvwdsetpudpf2uuwrck9chp8hyh/profile
Agentverse profile: Airlock Reader	https://agentverse.ai/agents/details/agent1qfta5ks9xw4ksj9jqhdhlyp8ulavnhg5agzkfdgep43g0mvgw23nz3ecqpx/profile
Agentverse profile: Airlock Bank	https://agentverse.ai/agents/details/agent1qwfmnsn4hykpj7k2ym2q4k8vq4fv3wry7hlyft8jewpec0ypn0eeqafqjpg/profile

2. Problem

Any AI agent that acts on your email is one well-worded message away from acting for an attacker: nothing in a normal build distinguishes 'the user asked for this' from 'an email told the agent to do this'. Prompt-level defences get bypassed by attackers who adapt to them. This is the exact shape of the real 'your bank details have changed, pay here instead' scam.

3. Target users

• Anyone who would let an agent pay routine bills but not hand it a blank cheque

• Older or busy people targeted by payment-redirect scams

• Fintech teams who want an autonomous payments agent they can defend

4. Solution and outcome

• Pays routine bills that fall inside limits you authored (handled silently)

• Holds anything suspicious as an approval card with the plain-English reason

• Settles an approved bill in FET via the Payment Protocol when you tap Approve

• Keeps a hash-chained audit log; ships a 7-attack red team with zero unauthorised effects

5. Mandatory requirements compliance

Requirement	Evidence
Agent registered on Agentverse	3 mailbox agents; profile links in section 1
Agent Chat Protocol (ACP)	AgentChatProtocol 0.3.0 with published manifest (agents/orchestrator/chat.py)
Discoverable & usable via ASI:One	Agentverse profile + README keywords; 'Chat with Agent'; shared chat link
Meaningful tool execution / orchestration	You (ASI:One) <-> Airlock orchestrator [Chat Protocol + Cards + Payment Protocol]. Reader agent supplies untrusted email text; the ASI:One LLM plans; a deterministic gate (never reads the email) decides ALLOW/ESCALATE/BLOCK; the Actor executes allowed actions; the Bank agent settles approved ones.
Workflow fully inside ASI:One	Request -> cards -> result -> action, no custom frontend
Public GitHub repo with run/test steps	README, .env.example, setup scripts, offline tests

6. Architecture

You (ASI:One) <-> Airlock orchestrator [Chat Protocol + Cards + Payment Protocol]. Reader agent supplies untrusted email text; the ASI:One LLM plans; a deterministic gate (never reads the email) decides

ALLOW/ESCALATE/BLOCK; the Actor executes allowed actions; the Bank agent settles approved ones.

Agent	Role	Address	Protocols
Airlock (orchestrator)	ASI:One-facing; owns the planner + deterministic gate + actor; chat, cards, payments	agent1qvlcnk524vntglwps897gf0dst0ugzcznvwdsetpudpf2uuwrck9chp8hyh	AgentChatProtocol 0.3.0, AgentPaymentProtocol 0.1.0 (seller)
Airlock Reader	The only component that touches raw untrusted content; parses inbox/email	agent1qfta5ks9xw4ksj9jqhdhlyp8ulavnhg5agzkfdgep43g0mvgw23nz3ecqpx	AirlockReader 1.0.0
Airlock Bank	Accepts only typed, gate-approved actions and settles them	agent1qwfmnsn4hykpj7k2ym2q4k8vq4fv3wry7hlyft8jewpec0ypn0eeqafqjpg	AirlockBank 1.0.0

7. Fetch.ai technology used

• uAgents + Agentverse mailbox for all three agents

• Agent Chat Protocol for ASI:One discoverability and interaction

• ASI:One LLM used as the planner that reads untrusted content (and is never trusted by the gate)

• ASI:One Interactive Cards: inbox summary, standing mandates, and an approve/decline review card per held decision

• Agent Payment Protocol (seller): approved bills settle on-chain in FET (testnet by default)

• Typed agent-to-agent protocols with send_and_receive, timeouts and local fallback

8. User workflow in ASI:One

[Screenshot placeholder: Run my inbox: two bills paid, two held; save as docs/screenshots/1_inbox_summary.png]

[Screenshot placeholder: An approval card with the plain-English reasons; save as docs/screenshots/2_review_card.png]

[Screenshot placeholder: Approving settles the bill in FET (Payment Protocol); save as docs/screenshots/3_payment.png]

[Screenshot placeholder: A direct injection email is held, not paid; save as docs/screenshots/4_injection_held.png]

[Screenshot placeholder: Red-team table: 7 attacks, 0 unauthorised effects; save as docs/screenshots/5_redteam.png]

9. Reliability and error handling

Failure	Behaviour
Planner (LLM) captured by an injection	The gate ignores the planner's reasoning; a captured planner changes nothing (attacks A1, A6)
Reader or Bank agent offline	Orchestrator falls back to doing the step locally; the user still gets a reply
ASI:One LLM down or no key	Planner falls back to a deterministic planner; the gate is unaffected
Payment unverifiable on-chain	CancelPayment; nothing is settled; the decision stays open
Guardrail flooding	The escalation queue deduplicates and, on overflow, locks down (stops acting, not asking)

10. Monetisation

• Approved bills settle in FET via the Payment Protocol (a settlement fee in production)

• The provenance-checked action boundary (Reader + gate) can be sold to other agents

• A subscription for households wanting autonomous, bounded bill payment

11. Impact

Autonomous bill-paying that is safe against the exact scam that targets real people. Seven adaptive attacks written against its own layers produce zero unauthorised effects, even when the planner is fully captured.

12. How to run / test

bash setup.sh (or setup.ps1) . add ASI1_API_KEY to .env . python scripts/demo_pipeline.py . python run_all.py . connect each agent's mailbox in the Agent Inspector

13. Limitations and roadmap

• Real inbox (IMAP/Gmail) and a real banking/UPI rail behind the Bank agent

• Per-user persisted store and mandates; a web console for mandates and the audit trail